Assignment 3.3 SSH Configuration and Hardening

Part 1:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ritup>ssh -o PreferredAuthentications=password -o PubkeyAuthentication=no pujari.ri@cy5001.mscy.org -p 20045
pujari.ri@cy5001.mscy.org's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-173-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Tue Mar 24 15:11:55 2026 from 71.233.186.221
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

pujari.ri@cy5001Lab3:~$ service ssh status
 * sshd is running
pujari.ri@cy5001Lab3:~$ netstat -ntlp
(No info could be read for "-p": geteuid()=1000 but you should be root.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      -
tcp6       0      0 :::22                   :::*                    LISTEN      -
pujari.ri@cy5001Lab3:~$
```

Figure 1: Login to the Linux VM using password-based authentication.

Part 2:

```
C:\Users\ritup>ssh-keygen -t ed25519 -o -a 100 -C "pujari.ri"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\ritup/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\ritup/.ssh/id_ed25519
Your public key has been saved in C:\Users\ritup/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:8TbBvnCXnK5TcfgcyT9J8mTfFNv9Nx3jF+5Eaku2mvA pujari.ri
The key's randomart image is:
+--[ED25519 256]--+
|                 |
|          .     . |
|         . o  o .=|
|          + oo+=X+|
|         S * ==@+X|
|          + =.=oOO|
|          ...= = =|
|           +..o . |
|            .E.   |
+----[SHA256]-----+

C:\Users\ritup>
```

Figure 2: Output of the ssh-keygen

```
pujari.ri@cy5001Lab3: ~        Command Prompt
pujari.ri@cy5001Lab3:~$ mkdir -p ~/.ssh
pujari.ri@cy5001Lab3:~$ chmod 700 ~/.ssh
pujari.ri@cy5001Lab3:~$ nano ~/.ssh/authorized_keys
pujari.ri@cy5001Lab3:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILIesmNfx9FJbOMLRrMauCE3BgGVgIrVIkq+EjnS4+IO pujari.ri
pujari.ri@cy5001Lab3:~$ chmod 600 ~/.ssh/authorized_keys
pujari.ri@cy5001Lab3:~$ ls -la ~/.ssh
total 12
drwx------ 2 pujari.ri pujari.ri 4096 Mar 24 15:52 .
drwxr-x--- 5 pujari.ri pujari.ri 4096 Mar 24 15:50 ..
-rw------- 1 pujari.ri pujari.ri   91 Mar 24 15:50 authorized_keys
pujari.ri@cy5001Lab3:~$
```

Figure 3: Permissions of the .ssh directory

```
pujari.ri@cy5001Lab3:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILIesmNfx9FJbOMLRrMauCE3BgGVgIrVIkq+EjnS4+IO pujari.ri
pujari.ri@cy5001Lab3:~$
```

Figure 4: Contents of the .ssh/authorized_keys file

```
C:\Users\ritup>type C:\Users\ritup\.ssh\id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILIesmNfx9FJbOMLRrMauCE3BgGVgIrVIkq+EjnS4+IO pujari.ri

C:\Users\ritup>ssh -i C:\Users\ritup\.ssh\id_ed25519 pujari.ri@cy5001.mscy.org -p 20045
Enter passphrase for key 'C:\Users\ritup\.ssh\id_ed25519':
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-173-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Tue Mar 24 15:34:13 2026 from 71.233.186.221
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

pujari.ri@cy5001Lab3:~$
```

Figure 5: Successful connection to the VM using public key authentication.

Part 3:

```
pujari.ri@cy5001Lab3:~$ sudo cat /var/log/auth.log | grep sshd | tail -20
Mar 25 15:07:15 cy5001Lab3 sshd[358]: Starting session: shell on pts/1 for pujari.ri from 73.68.53.135 port 6654 id 0
Mar 25 15:07:18 cy5001Lab3 sshd[347]: syslogin_perform_logout: logout() returned an error
Mar 25 15:07:18 cy5001Lab3 sshd[358]: Received disconnect from 73.68.53.135 port 6654:11: disconnected by user
Mar 25 15:07:18 cy5001Lab3 sshd[358]: Disconnected from user pujari.ri 73.68.53.135 port 6654
Mar 25 15:07:18 cy5001Lab3 sshd[347]: pam_unix(sshd:session): session closed for user pujari.ri
Mar 25 15:07:20 cy5001Lab3 sshd[363]: Connection from 73.68.53.135 port 6649 on 172.17.0.31 port 22 rdomain ""
Mar 25 15:07:20 cy5001Lab3 sshd[363]: Accepted key ED25519 SHA256:8TbBvnCXnK5TcfgcyT9J8mTfFNv9Nx3jF+5Eaku2mvA found at /home/pujari.r
i/.ssh/authorized_keys:1
Mar 25 15:07:20 cy5001Lab3 sshd[363]: Postponed publickey for pujari.ri from 73.68.53.135 port 6649 ssh2 [preauth]
Mar 25 15:07:25 cy5001Lab3 sshd[363]: Accepted key ED25519 SHA256:8TbBvnCXnK5TcfgcyT9J8mTfFNv9Nx3jF+5Eaku2mvA found at /home/pujari.r
i/.ssh/authorized_keys:1
Mar 25 15:07:25 cy5001Lab3 sshd[363]: Accepted publickey for pujari.ri from 73.68.53.135 port 6649 ssh2: ED25519 SHA256:8TbBvnCXnK5Tc
fgcyT9J8mTfFNv9Nx3jF+5Eaku2mvA
Mar 25 15:07:25 cy5001Lab3 sshd[363]: pam_unix(sshd:session): session opened for user pujari.ri(uid=1000) by (uid=0)
Mar 25 15:07:25 cy5001Lab3 sshd[363]: pam_systemd(sshd:session): Failed to connect to system bus: No such file or directory
Mar 25 15:07:25 cy5001Lab3 sshd[363]: pam_limits(sshd:session): Could not set limit for 'core' to soft=-1, hard=-1: Operation not per
mitted; uid=0,euid=0
Mar 25 15:07:25 cy5001Lab3 sshd[363]: pam_env(sshd:session): Unable to open env file: /etc/default/locale: No such file or directory
Mar 25 15:07:25 cy5001Lab3 sshd[363]: User child is on pid 374
Mar 25 15:07:25 cy5001Lab3 sshd[374]: Starting session: shell on pts/1 for pujari.ri from 73.68.53.135 port 6649 id 0
Mar 25 15:07:28 cy5001Lab3 sshd[363]: syslogin_perform_logout: logout() returned an error
Mar 25 15:07:28 cy5001Lab3 sshd[374]: Received disconnect from 73.68.53.135 port 6649:11: disconnected by user
Mar 25 15:07:28 cy5001Lab3 sshd[374]: Disconnected from user pujari.ri 73.68.53.135 port 6649
Mar 25 15:07:28 cy5001Lab3 sshd[363]: pam_unix(sshd:session): session closed for user pujari.ri
pujari.ri@cy5001Lab3:~$
```

```
Mar 26 02:28:10 cy5001Lab3 sshd[523]: pam_unix(sshd:session): session opened for user pujari.ri(uid=1000) by (uid=0)
Mar 26 02:28:10 cy5001Lab3 sshd[523]: pam_systemd(sshd:session): Failed to connect to system bus: No such file or directory
Mar 26 02:28:10 cy5001Lab3 sshd[523]: pam_limits(sshd:session): Could not set limit for 'core' to soft=-1, hard=-1: Operation not per
mitted; uid=0,euid=0
Mar 26 02:28:10 cy5001Lab3 sshd[523]: pam_env(sshd:session): Unable to open env file: /etc/default/locale: No such file or directory
Mar 26 02:28:10 cy5001Lab3 sshd[523]: User child is on pid 534
Mar 26 02:28:10 cy5001Lab3 sshd[534]: Starting session: shell on pts/1 for pujari.ri from 71.233.187.16 port 65272 id 0
Mar 26 02:30:59 cy5001Lab3 sshd[523]: syslogin_perform_logout: logout() returned an error
Mar 26 02:30:59 cy5001Lab3 sshd[534]: Received disconnect from 71.233.187.16 port 65272:11: disconnected by user
Mar 26 02:30:59 cy5001Lab3 sshd[534]: Disconnected from user pujari.ri 71.233.187.16 port 65272
Mar 26 02:30:59 cy5001Lab3 sshd[523]: pam_unix(sshd:session): session closed for user pujari.ri
Mar 26 02:31:12 cy5001Lab3 sshd[544]: Connection from 71.233.187.16 port 65309 on 172.17.0.31 port 22 rdomain ""
Mar 26 02:31:12 cy5001Lab3 sshd[544]: Invalid user attacker from 71.233.187.16 port 65309
Mar 26 02:31:12 cy5001Lab3 sshd[544]: Connection reset by invalid user attacker 71.233.187.16 port 65309 [preauth]
Mar 26 02:31:15 cy5001Lab3 sshd[546]: Connection from 71.233.187.16 port 65284 on 172.17.0.31 port 22 rdomain ""
Mar 26 02:31:15 cy5001Lab3 sshd[546]: Invalid user attacker from 71.233.187.16 port 65284
Mar 26 02:31:15 cy5001Lab3 sshd[546]: Connection reset by invalid user attacker 71.233.187.16 port 65284 [preauth]
Mar 26 02:31:21 cy5001Lab3 sshd[548]: Connection from 71.233.187.16 port 65270 on 172.17.0.31 port 22 rdomain ""
Mar 26 02:31:21 cy5001Lab3 sshd[548]: Invalid user attacker from 71.233.187.16 port 65270
Mar 26 02:31:21 cy5001Lab3 sshd[548]: Connection reset by invalid user attacker 71.233.187.16 port 65270 [preauth]
Mar 26 02:31:29 cy5001Lab3 sshd[550]: Connection from 71.233.187.16 port 65167 on 172.17.0.31 port 22 rdomain ""
Mar 26 02:31:29 cy5001Lab3 sshd[550]: Accepted key ED25519 SHA256:8TbBvnCXnK5TcfgcyT9J8mTfFNv9Nx3jF+5Eaku2mvA found at /home/pujari.r
i/.ssh/authorized_keys:1
Mar 26 02:31:29 cy5001Lab3 sshd[550]: Postponed publickey for pujari.ri from 71.233.187.16 port 65167 ssh2 [preauth]
Mar 26 02:31:34 cy5001Lab3 sshd[550]: Accepted key ED25519 SHA256:8TbBvnCXnK5TcfgcyT9J8mTfFNv9Nx3jF+5Eaku2mvA found at /home/pujari.r
i/.ssh/authorized_keys:1
Mar 26 02:31:34 cy5001Lab3 sshd[550]: Accepted publickey for pujari.ri from 71.233.187.16 port 65167 ssh2: ED25519 SHA256:8TbBvnCXnK5
TcfgcyT9J8mTfFNv9Nx3jF+5Eaku2mvA
Mar 26 02:31:34 cy5001Lab3 sshd[550]: pam_unix(sshd:session): session opened for user pujari.ri(uid=1000) by (uid=0)
Mar 26 02:31:34 cy5001Lab3 sshd[550]: pam_systemd(sshd:session): Failed to connect to system bus: No such file or directory
Mar 26 02:31:34 cy5001Lab3 sshd[550]: pam_limits(sshd:session): Could not set limit for 'core' to soft=-1, hard=-1: Operation not per
mitted; uid=0,euid=0
Mar 26 02:31:34 cy5001Lab3 sshd[550]: pam_env(sshd:session): Unable to open env file: /etc/default/locale: No such file or directory
Mar 26 02:31:34 cy5001Lab3 sshd[550]: User child is on pid 561
Mar 26 02:31:34 cy5001Lab3 sshd[561]: Starting session: shell on pts/1 for pujari.ri from 71.233.187.16 port 65167 id 0
pujari.ri@cy5001Lab3:~$
```

Figure 6: Logs from your last connections

```
C:\Users\ritup>ssh pujari.ri@cy5001.mscy.org -p 20045
Ubuntu 22.04.5 LTS
 ***************************************************************************
                            NOTICE TO CY5001 USERS

    This system is part of the CY5001 infrastructure. It is for authorized
    use only. Users (authorized or unauthorized) have no explicit or implicit
    expectation of privacy.

 ***************************************************************************
Enter passphrase for key 'C:\Users\ritup/.ssh/id_ed25519':
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-173-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Thu Mar 26 02:31:34 2026 from 71.233.187.16
pujari.ri@cy5001Lab3:~$
```

Figure 7: Showing the banner of the SSH connection attempt.

Part 4:

Classify the security controls you put in practice.

Preventive controls:

1. **Disable Password Authentication (PasswordAuthentication no):** By completely removing the password-based login, it prevents brute force attacks.
2. **Public Key Authentication (Part 2):** This method allows authentication only by those users with valid private keys and eliminates all other users from being able to authenticate.
3. **Use PermitRootLogin no (Part 3):** This prevents attackers from being able to directly target the most privileged account via SSH. Even if a user account is compromised, the attacker will not immediately have access to / be able to take control of the entire system.
4. **Configuring the Welcome Banner (Part 3):** This will assist as a deterrent as well as serve as a preventive measure by clearly stating acceptable use of the systems. It also helps to discourage casual intruders; while simultaneously providing a legally defined "No Trespassing" sign required for prosecution, which could also potentially prevent some types of unauthorized activity.
5. **Use Proper Permissions on .ssh (700) and authorized_keys (600) Files (Part 2):** Having proper file permissions on the authentication keys will ensure that any

unauthorized user cannot read or modify authentication keys. Preventing against local privilege escalation.

6. **Disable KbdInteractiveAuthentication (Part 3):** This removes another means of authentication that an attacker could exploit.

Detective Controls:

1. **LogLevel VERBOSE (Part 3):** Provides maximum detail for all SSH connection attempts, and provides the ability to detect unauthorized/suspicious access.
2. **Monitor /var/log/auth.log (Part 3):** Provides visibility into SSH connections that have successfully connected, failed to connect, and allows for detection of potentially malicious activity associated.
3. **Run rsyslogd (Part 3):** Ensures that system security logs associated are continuously collected and stored for monitoring.

Corrective Controls:

1. **Backup Configuration (Project 1):** The backup configuration that was reviewed and updated during Project 1 is considered corrective control. It allows for the restoration of data and system state after a security incident has occurred. Therefore, it minimizes damage caused by the security incident and the recovery of the incident.
2. **Hard drive encryption (Project 1):** While encryption has preventive aspects; the primary corrective value of hard drive encryption is the prevention of data access by attackers after hardware has been lost/stolen or compromised. Therefore, hard drive encryption will minimize any damages that occurred as a result of the physical security incident.
3. **Antivirus Malware Remediation (Project 1):** When antivirus software identifies malware and subsequently quarantines or deletes the file from the system, it is acting as a corrective control. The antivirus software has prevented an active security issue by removing the threat after it has been detected.